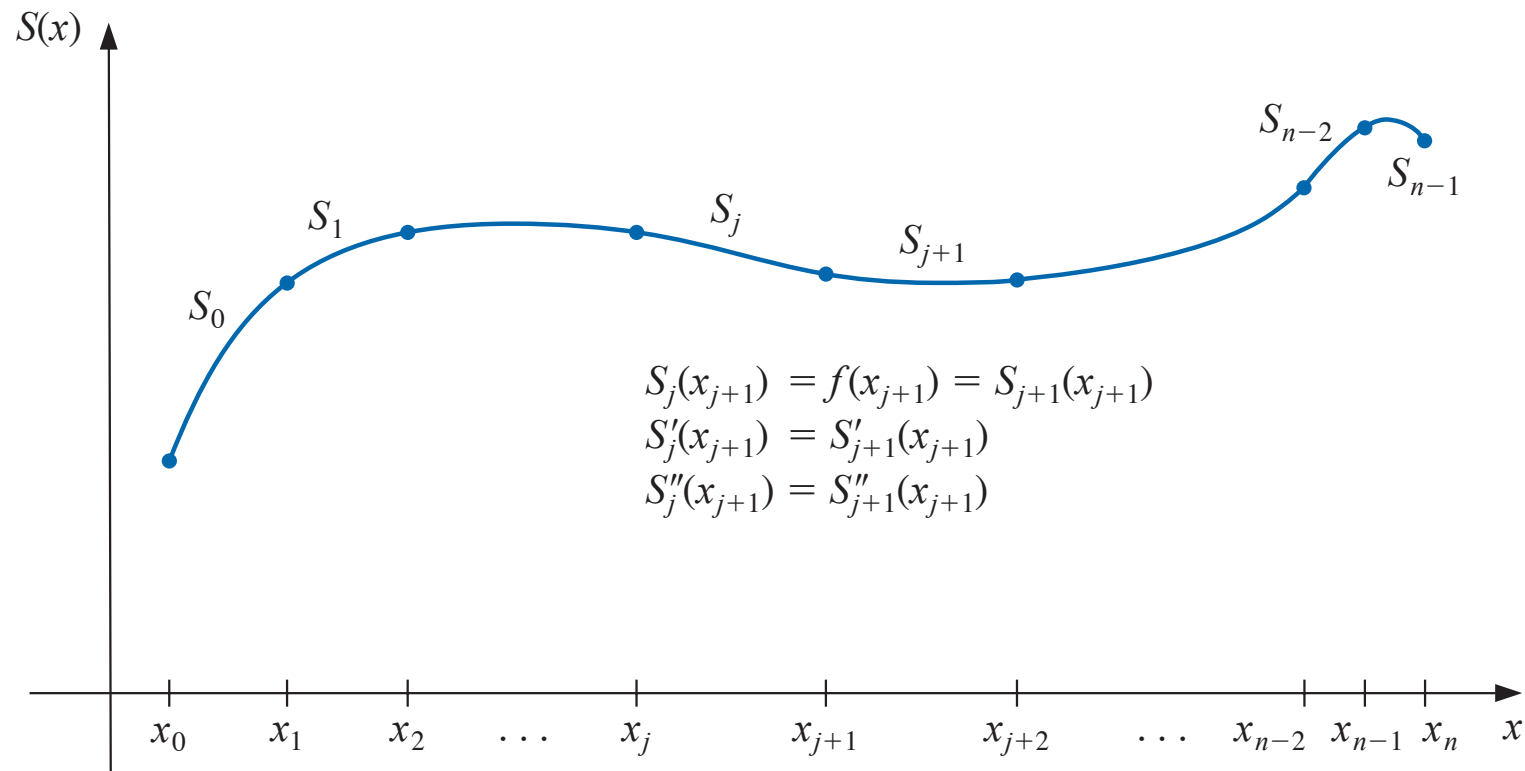


Cubic spline interpolation

High-degree polynomial fitting has strong oscillations.

Can we get a piecewise “low degree” polynomial interpolation instead?



Cubic spline interpolation

Suppose we are given $x_0, \dots, x_n$ and $f(x_i)$, $\forall i$, we want to find a cubic spline interpolation $S(x)$, s.t.

1. $S(x)$ is a cubic polynomial, denoted by $S_j(x)$, on $[x_j, x_{j+1}]$;
2. $S_j(x_j) = f(x_j)$, $S_j(x_{j+1}) = f(x_{j+1})$
3. $S_j(x_j) = S_{j+1}(x_j)$ for all j (consequence of Item 2.)
4. $S'_{j+1}(x_{j+1}) = S'_j(x_{j+1})$ for all j
5. $S''_{j+1}(x_{j+1}) = S''_j(x_{j+1})$ for all j
6. One of the following boundary condition is satisfied:
 - ▶ $S''(x_0) = S''(x_n) = 0$ (natural/free boundary condition)
 - ▶ $S'(x_0) = f'(x_0)$ and $S'(x_n) = f'(x_n)$ (clamped boundary condition)

Cubic spline interpolation

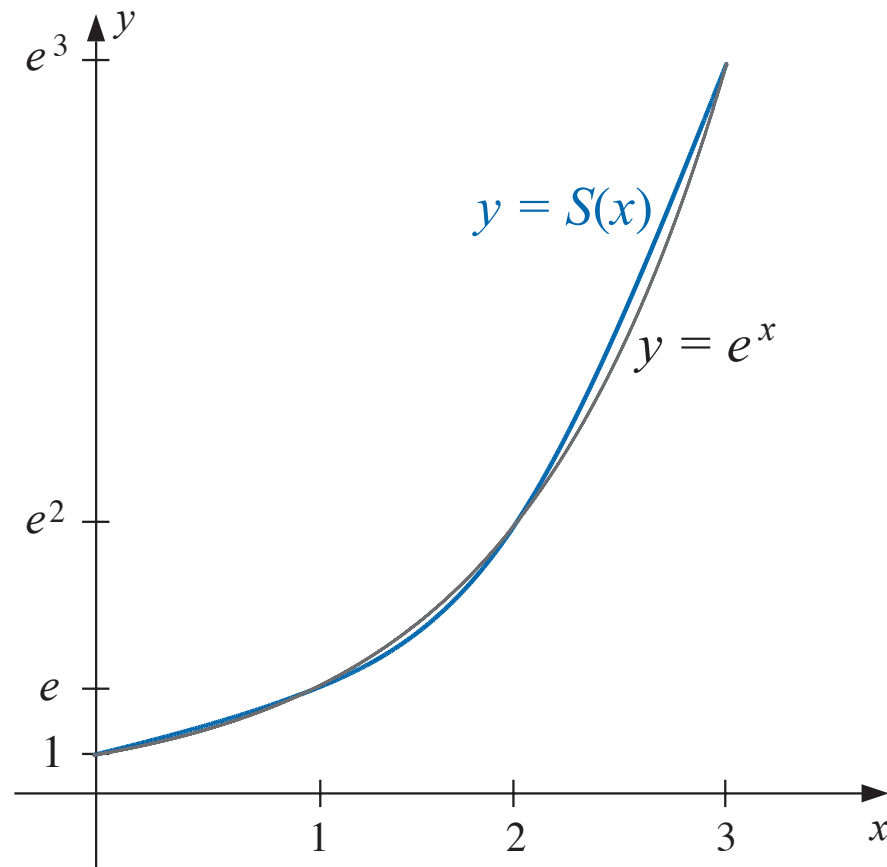
Remarks

1. $S(x)$ only agrees with $f(x)$ at x_i , not necessarily $f'(x)$.
2. Clamped BC is more accurate than natural BC, but needs $f'(a), f'(b)$.

Example

Example (Construct a natural cubic spline)

Construct natural cubic spline for $f(x) = e^x$ using $x_i = i$ for $i = 0, 1, 2, 3$.



Construction of cubic splines

Given $\{(x_i, f(x_i)) : i = 0, \dots, n\}$, we need to construct n cubic polynomials, each with 4 coefficients

$$S_j(x) = a_j + b_j(x - x_j) + c_j(x - x_j)^2 + d_j(x - x_j)^3 \text{ on } [x_j, x_{j+1}], \quad \forall j$$

So we have $4n$ unknowns to determine:

$$a_j, b_j, c_j, d_j, \quad \text{for } j = 0, \dots, n - 1$$

Construction of cubic splines

The cubic spline conditions will determine these $4n$ coefficients uniquely ($h_j := x_{j+1} - x_j$) according to the 6 rules:

1. By definition of S_j .
2. Since $S_j(x_j) = a_j = f(x_j)$, we get a_j for $j = 0, \dots, n-1$.
3. $a_{j+1} = S_{j+1}(x_{j+1}) = S_j(x_{j+1}) = a_j + b_j h_j + c_j h_j^2 + d_j h_j^3$.
4. $S'_j(x) = b_j + 2c_j(x - x_j) + 3d_j(x - x_j)^2$, therefore $S'_j(x_j) = b_j$ and $b_{j+1} = S'_{j+1}(x_{j+1}) = S'_j(x_{j+1}) = b_j + 2c_j h_j + 3d_j h_j^2$.
5. $S''_j(x) = 2c_j + 6d_j(x - x_j)$. Then we have $S''_j(x_j) = 2c_j$. So $2c_{j+1} = S''_{j+1}(x_{j+1}) = S''_j(x_j) = 2c_j + 6d_j h_j$.
6. Use the required boundary condition.
 - ▶ Natural BC: $c_0 = c_n = 0$
 - ▶ Clamped BC: $b_0 = f'(a)$, $b_n = f'(b)$.

Construction of cubic splines

As we have known the values of a_j , we can combine equations from the last 3 items to solve for c_j and obtain

$$h_{j-1}c_{j-1} + 2(h_{j-1} + h_j)c_j + h_jc_{j+1} = \frac{3}{h_j}(a_{j+1} - a_j) - \frac{3}{h_{j-1}}(a_j - a_{j-1})$$

for each $j = 1, \dots, n-1$. If we assume natural splines with $S''_0(x_0) = S''_{n-1}(x_n) = 0$, then $c_0 = c_n = 0$.